

CISA Manufacturing OT Ransomware Tabletop

A discussion-based incident-response tabletop plan for a fictional auto-parts manufacturer, informed by CISA Critical Manufacturing CTEP materials.

Status: scenario and facilitator materials prepared; exercise conduct not evidenced

Executive summary

Prepared a 60–75 minute scenario for Northstar Components, a fictional Ontario auto-parts manufacturer. Injects progress from rapid file changes on an engineering workstation to engineering-share disruption, SMB activity toward OT, a third-party remote-support question, and recovery uncertainty. The scenario tests coordination and safe decision-making while separating observations from assumptions.

Scope and method

Exercise design only: facilitator guide, role and decision prompts, fictional evidence, and after-action template. No participants, real plant, vendor, equipment, or operational system are documented as having taken part.

Results

Exercise objective	What the facilitator should observe
Coordination	Named incident lead, scribe, severity decision, update cadence, and escalation owner.
OT-aware response	Participants identify operations and safety decision owners and distinguish host, network, and operator reports.
Safe containment	Any proposed account, host, or network action has an authorized approver and an impact check.
Recovery readiness	Known-good backup validation, clean credentials, process checks, and recovery gates are identified.
After-action	Specific improvement actions have one owner, target date, and completion evidence.

Evidence screenshots

Figure 1. CTEP source citation and report build

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- shlink source-to-destination summary
Terminal Critical Manufacturing CTEP Situation Manual

## Source

Cybersecurity and Infrastructure Security Agency (CISA), *Critical Manufacturing CTEP Situation Manual*, January 2024. Scenario material adapted for this fictional, independent tabletop walkthrough.

akachi@akachi-ThinkPad-T440s:~$ command -v libreoffice
/snap/bin/libreoffice
akachi@akachi-ThinkPad-T440s:~$ command -v pandoc
/usr/bin/pandoc
akachi@akachi-ThinkPad-T440s:~$ cd ~/ot-cisa-tabletop && pandoc REPORT.md -o REPORT.docx && libreoffice --headless --convert-to pdf --outdir . REPORT.docx
libxbackend-1.0.so: cannot open shared object file: No such file or directory
Failed to load module: /home/akachi/snap/libreoffice/common/cache/gio-modules/libgiolibproxy.so
Warning: Schema "org.gnome.system.locale" has path "/system/locale/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
Warning: Schema "org.gnome.system.proxy" has path "/system/proxy/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
Warning: Schema "org.gnome.system.proxy.http" has path "/system/proxy/http/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
Warning: Schema "org.gnome.system.proxy.https" has path "/system/proxy/https/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
Warning: Schema "org.gnome.system.proxy.ftp" has path "/system/proxy/ftp/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
Warning: Schema "org.gnome.system.proxy.socks" has path "/system/proxy/socks/". Paths starting with "/apps/", "/desktop/" or "/system/" are deprecated.
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Terminal excerpt from the tabletop preparation folder showing the CISA Critical Manufacturing CTEP source citation and report-preparation attempt. It is evidence of scenario preparation only; no conducted exercise is evidenced.

Figure 2. CISA CTEP inject timeline, evidence plan, and exercise questions

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Organization: Northstar Components (fictional manufacturer)
Exercise basis: CISA Critical Manufacturing CTEP Situation Manual
Scope: Discussion-based tabletop; no live plant or PLC testing

## Scenario timeline

| Day | Exercise inject |
|---|---|
| 1 | Advisory warns of a threat group targeting manufacturers, with IT compromise, data theft, persistence, and destructive malware affecting OT. |
| 5 | Staff request urgent installation of new production software. |
| 7 | A vendor arrives without notice to update legacy IT and OT systems. |
| 9 | Employees receive a vendor-themed email offering free event tickets. |
| 15 | The vendor confirms its email account was compromised and the ticket link was malicious. |
| 20 | A researcher reports company trade secrets advertised online; the insider risk team confirms data exfiltration. |
| 22, morning | Shared files are missing or altered. Backups restore files, but recent work is lost. |
| 22, afternoon | Internet-facing PLCs malfunction; PLC data is encrypted and IT/OT staff are locked out of their accounts. |
| 29 | PLC access remains unavailable. The plant is operating manually; an online post claims a known exploited vulnerability was used. |

## Evidence plan

- Suricata: use the GridWatch lab alert output as simulated OT network evidence.
- Zeek: use GridWatch connection metadata as simulated network-flow evidence.
- Splunk: add a searchable incident timeline after importing or creating clearly labeled lab events.
- Keep the CISA scenario injects separate from observed tool output; the CISA manual is an exercise scenario, not telemetry.

## Exercise questions

1. Who can authorize stopping or restarting production?
2. How will IT and OT coordinate containment while protecting personnel and equipment?
3. What evidence must be preserved before systems are restored?
4. How will the plant verify PLC integrity before reconnecting or resuming operations?
5. What manual operating procedures and safety checks apply during the outage?
6. Who handles vendor coordination, legal review, and external notifications?
7. What recovery gaps and corrective actions should be recorded?

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Scenario notes listing the CISA Critical Manufacturing CTEP injects (Day 1 to Day 29) used as source material, an evidence plan that keeps lab tool output separate from scenario injects, and seven facilitator questions. Preparation material only; no participant responses are recorded. Terminal capture, 15:19.

Evidence screenshots (continued)

Figure 3. Facilitator decision prompts and evidence checklist

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- Who has authority to declare an OT incident and approve operational changes?
- Which safety conditions must be met before isolating affected systems?
- How will the organization verify PLC logic and configuration integrity before reconnection?
- What manual procedures, staffing, and safety checks apply during an outage?
- How will vendor access and emergency changes be authorized and documented?
- Who coordinates legal review, leadership updates, customers, and external notifications?
- Which backup, recovery, and evidence-preservation gaps require follow-up?

## Conclusion

The tabletop highlights the need for a joint II/OT response that accounts for safety, production, vendor access, evidence preservation, communications, and careful restoration. The Splunk and GridWatch work provides a separate, reproducible demonstration of searching simulated Modbus alert data.

## Evidence checklist

- GridWatch PCAP and Suricata 'eve.json'
- Suricata 'fast.log'
- Zeek 'conn.log' and 'packet_filter.log'
- Splunk upload-success screen
- Splunk search showing 122 events
- Splunk alert-signature count table
- Splunk source-to-destination summary
- CISA Critical Manufacturing CTEP Situation Manual
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Draft facilitator prompts covering OT incident authority, safety conditions before isolation, PLC integrity verification, manual operations, vendor access, and recovery gaps, followed by the evidence checklist. Terminal capture, 16:26.

CISA Manufacturing OT Ransomware Tabletop | Analyst notes

Recommended follow-up

1. Facilitate the fictional inject sequence and record decisions, approvers, rationale, and reassessment times.
2. Label statements as observed, reported, hypothesis, or decision.
3. Do not run commands, isolate systems, stop production, or contact organizations during the discussion.
4. Complete the after-action record with actual participant observations; do not invent MTDD, MTTR, or readiness outcomes.

Limitations

This is an exercise plan, not evidence a tabletop was conducted. It does not establish response readiness, compliance, real detection performance, or plant safety. All systems and evidence are fictional; operational use requires adaptation and approval by the organization.

Evidence basis

Local source material includes the CISA Critical Manufacturing CTEP Situation Manual (January 2024) and a prepared fictional facilitator guide. No completed participant decision log or after-action results were provided.

Skills demonstrated

OT incident-response planning, tabletop facilitation design, safety-conscious escalation, evidence discipline, and after-action tracking.

Portfolio integrity note: This report describes an educational demonstration. It is not a client engagement, production security assessment, or certification of control effectiveness.